

```

root_path = fileparts(mfilename("fullpath"));
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit200', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
used_ds = containers.Map()

```

```
used_ds =
```

```
Map with properties:
```

```

    Count: 0
   KeyType: char
  ValueType: any

```

```

sel_info = struct('cuts', [], 'en_range', [], 'fname', '', 'bg_par', []);
%
sel_info.cuts = cuts2fit200.cuts_dir001FFfit(1);
sel_info.en_range = 30:10:110;
sel_info.fname='EnFit_Ei200ref001off110_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br110dir001')=sel_info;
%
sel_info.cuts = cuts2fit200.cuts_dir001FFfit(2);
sel_info.en_range = 30:10:130;
sel_info.fname='EnFit_Ei200ref001off200_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br200dir001')=sel_info;

```

```

%cuts_list = cuts2fit200.cutsS_list();
%cuts_list = {cuts2fit200.cut200_200sel};
%cuts_list = {cuts2fit200.cut110_sel};
%cuts_list = {cut(cuts2fit200.cut200_sel,[0,0.02,1],[])};
select= 'Ei200Br200dir001';
sel_info = used_ds(select);

cuts_list = sel_info.cuts

```

```

cuts_list = 1x1 cell array
    {1x1 sqw}

```

```

mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
end

dir_name = "GH";

```

```

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning
%cut_en = 30:10:120;
%cut_en = 40:10:150;
%cut_en = 145;
cut_en = sel_info.en_range;

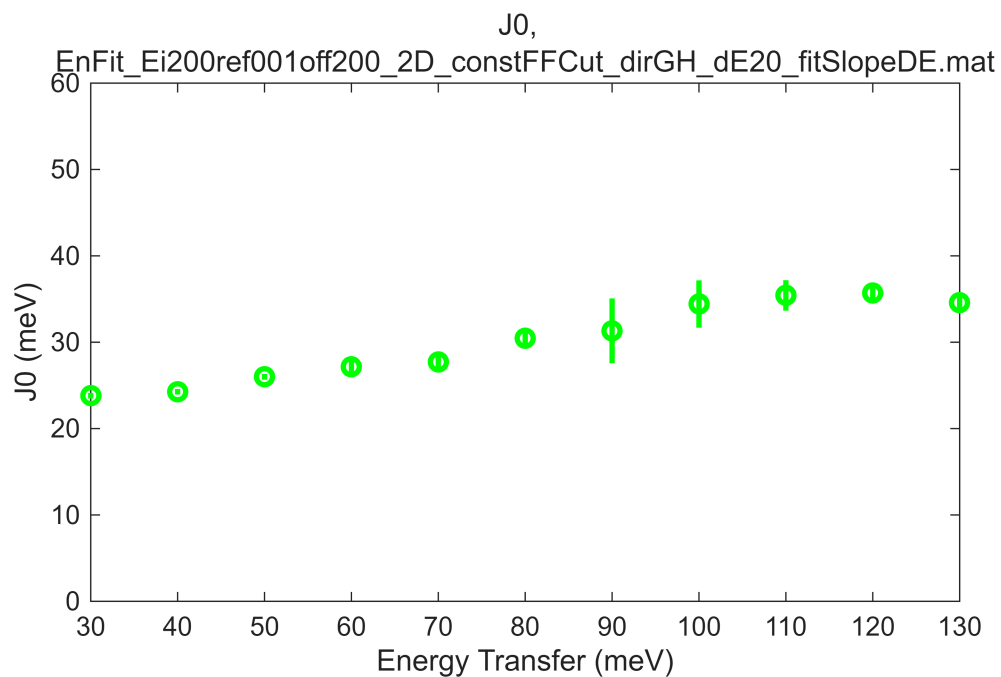
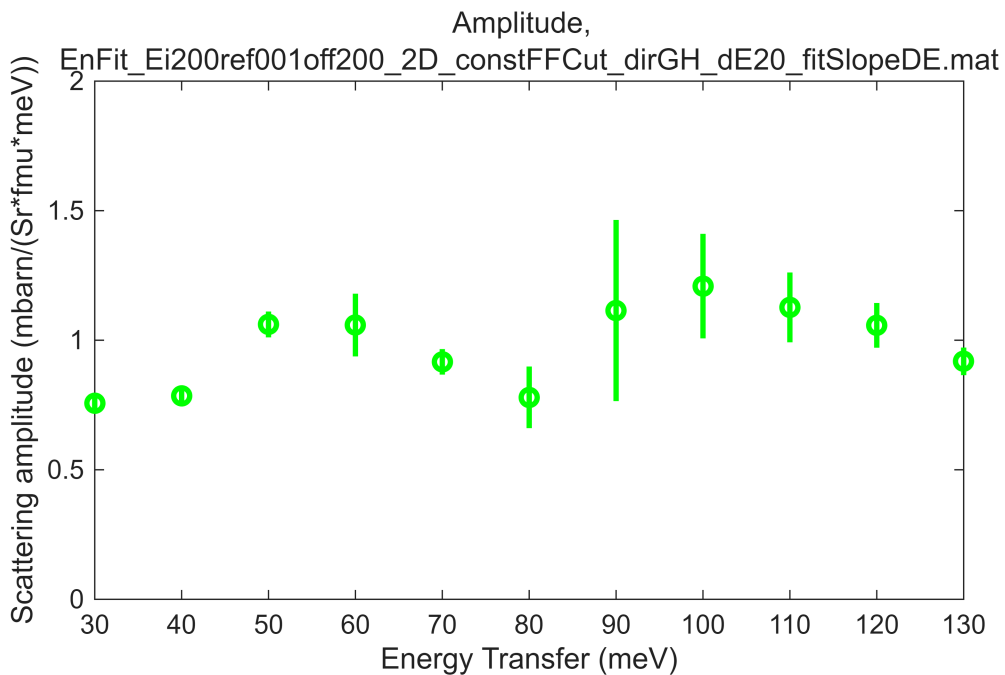
bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;

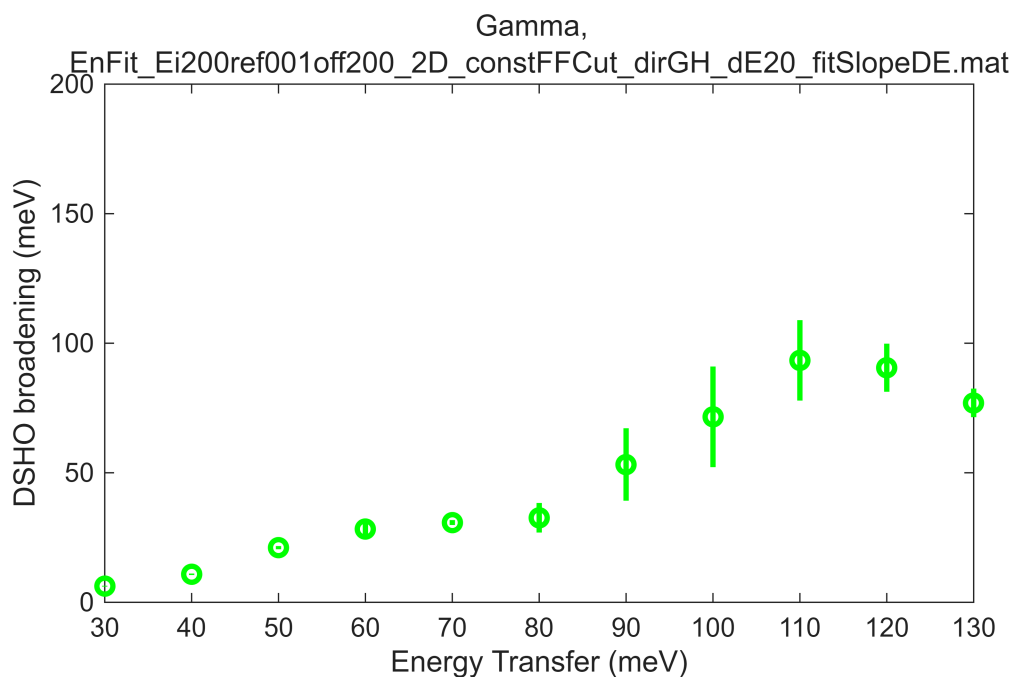
```

```

[fit_res_200fix_bg,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_en,dE_step,half_dE,true);

```





**** En = 30 ± 10

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.29326

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 99566 pixels -- processing data... -----> included 91532 pixels

Number of points in input object: 99566

Fraction of object selected: 100.00% (= 99566 points) ; Access speed: 0.26GB/sec

Fraction of object retained: 91.93% (= 91532 points)

**** Total time in cut(sqw) : 0.2sec Including:

Data access time fraction: 10.9%; Transf. time frac.: 87.5%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 99562 pixels -- processing data... -----> retained 91528 pixels

Number of points in input object: 99562

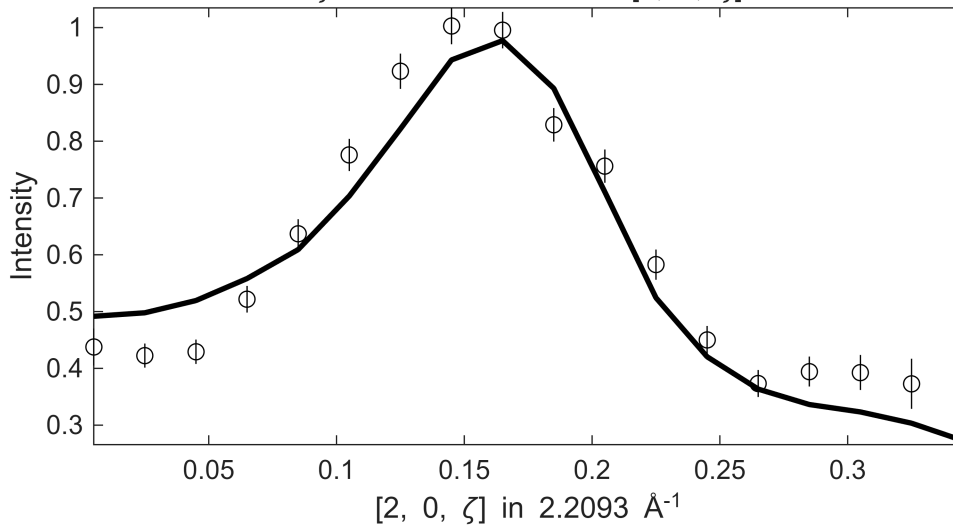
Fraction of object selected: 100.00% (= 99562 points) ; Access speed: 0.40GB/sec

Fraction of object retained: 91.93% (= 91528 points)

**** Total time in cut(sqw) : 0.2sec Including:

Data access time fraction: 9.0%; Transf. time frac.: 78.5%; Resulting pix preparation: 11.8%

Fe_ei200_align_sym3.sqw
 $S=0.74\pm0.081$; $J_0=24\pm0.38$; $\gamma=6.02511\pm1.21398$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $20 \leq E \leq 40$
 $\zeta=-0.005:0.02:0.615$ in $[2, 0, \zeta]$

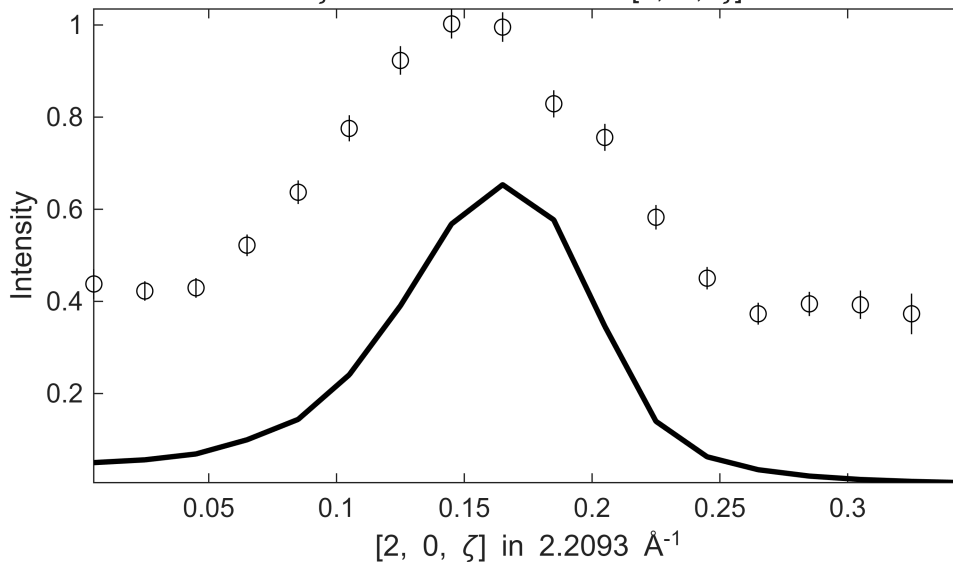


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 99566 pixels -- processing data... -----> retained 91532 pixels

Number of points in input object: 99566
 Fraction of object selected: 100.00% (= 99566 points) ; Access speed: 0.28GB/sec
 Fraction of object retained: 91.93% (= 91532 points)

**** Total time in cut(sqw) : 0.2sec Including:
 Data access time fraction: 11.4%; Transf. time frac.: 75.7%; Resulting pix preparation: 11.9%

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $20 \leq E \leq 40$
 $\zeta=-0.005:0.02:0.615$ in $[2, 0, \zeta]$



 **** $E_n = 40\pm10$

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.58651

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 116225 pixels -- processing data... -----> included 106019 pixels

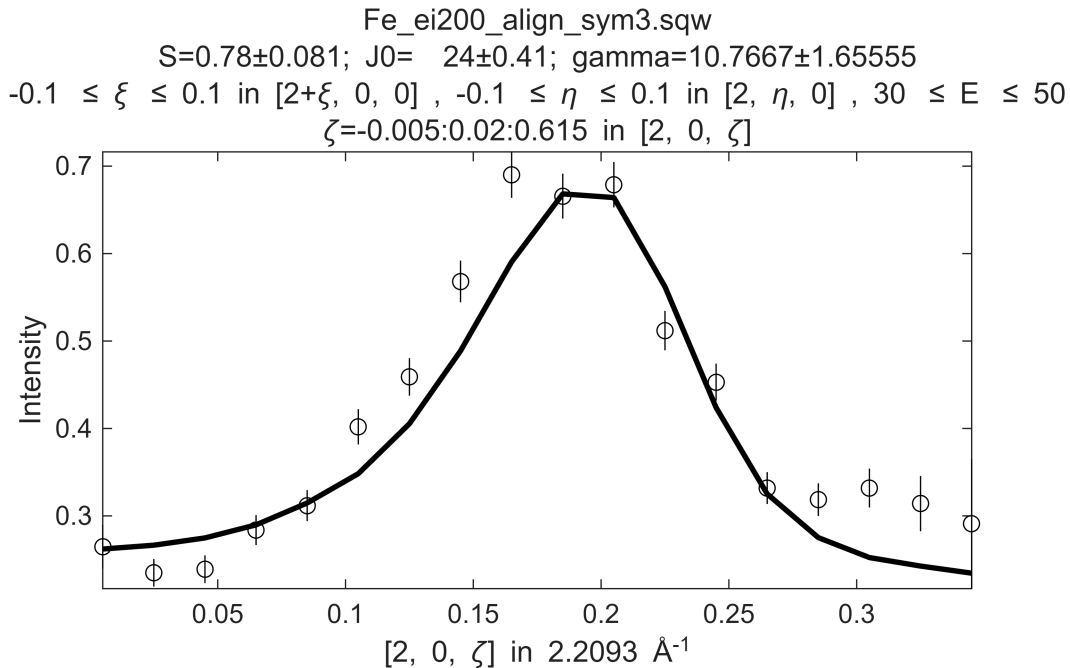
Number of points in input object: 116225
 Fraction of object selected: 100.00% (= 116225 points) ; Access speed: 1.94GB/sec
 Fraction of object retained: 91.22% (= 106019 points)

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 29.5%; Transf. time frac.: 57.0%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 116211 pixels -- processing data... -----> retained 106009 pixels
-----
Number of points in input object: 116211
      Fraction of object selected: 100.00% (= 116211 points) ; Access speed: 1.93GB/sec
      Fraction of object retained:  91.22% (= 106009 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 11.5%; Transf. time frac.: 57.9%; Resulting pix preparation: 27.5%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 116225 pixels -- processing data... -----> retained 106019 pixels
-----
Number of points in input object: 116225
      Fraction of object selected: 100.00% (= 116225 points) ; Access speed: 2.21GB/sec
      Fraction of object retained:  91.22% (= 106019 points)

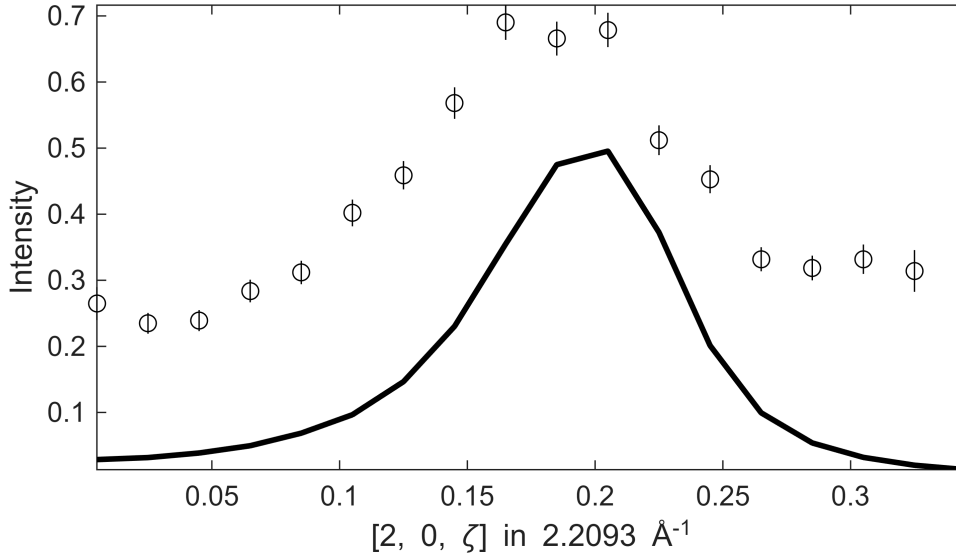
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 18.5%; Transf. time frac.: 62.3%; Resulting pix preparation: 19.1%
-----

```

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $30 \leq E \leq 50$

$\zeta = -0.005:0.02:0.615$ in $[2, 0, \zeta]$



**** En = 50±10

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.58651

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 120049 pixels -- processing data... -----> included 108793 pixels

Number of points in input object: 120049

Fraction of object selected: 100.00% (= 120049 points) ; Access speed: 2.28GB/sec

Fraction of object retained: 90.62% (= 108793 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 34.5%; Transf. time frac.: 58.4%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 120037 pixels -- processing data... -----> retained 108781 pixels

Number of points in input object: 120037

Fraction of object selected: 100.00% (= 120037 points) ; Access speed: 2.06GB/sec

Fraction of object retained: 90.62% (= 108781 points)

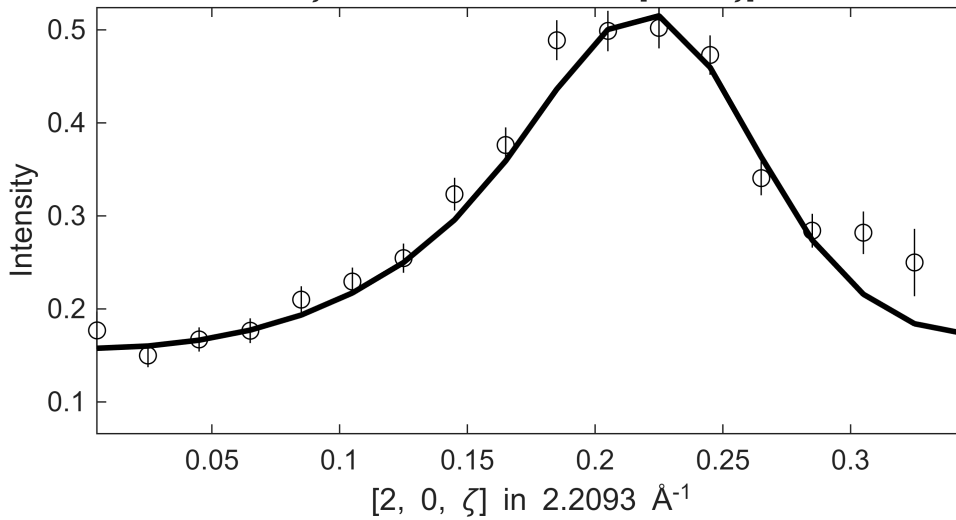
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 20.5%; Transf. time frac.: 57.4%; Resulting pix preparation: 22.1%

Fe_ei200_align_sym3.sqw

S= 1 ± 0.11 ; J0= 26 ± 0.55 ; gamma= 20.4597 ± 2.71175

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $40 \leq E \leq 60$
 $\zeta = -0.005:0.02:0.615$ in $[2, 0, \zeta]$

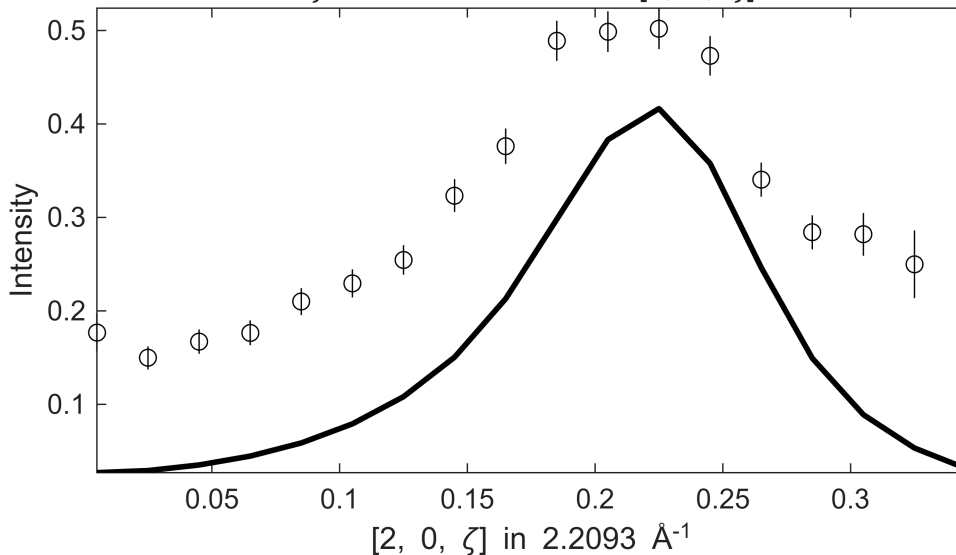


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 120049 pixels -- processing data... -----> retained 108793 pixels
-----
Number of points in input object: 120049
Fraction of object selected: 100.00% (= 120049 points) ; Access speed: 1.67GB/sec
Fraction of object retained: 90.62% (= 108793 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 23.2%; Transf. time frac.: 54.9%; Resulting pix preparation: 21.9%
-----
```

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $40 \leq E \leq 60$
 $\zeta = -0.005:0.02:0.615$ in $[2, 0, \zeta]$



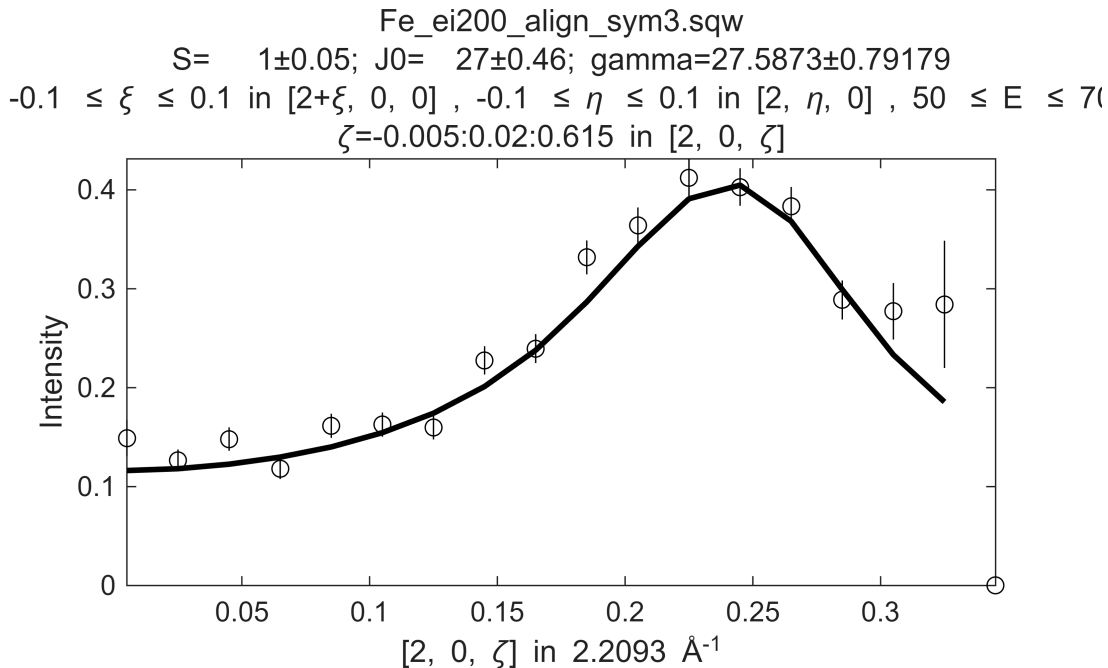
```
*****
**** En = 60±10
*****
Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.87977
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 124799 pixels -- processing data... -----> included 113303 pixels
-----
Number of points in input object: 124799
Fraction of object selected: 100.00% (= 124799 points) ; Access speed: 1.96GB/sec
Fraction of object retained: 90.79% (= 113303 points)
```

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 33.2%; Transf. time frac.: 60.3%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 124770 pixels -- processing data... -----> retained 113280 pixels
-----
Number of points in input object: 124770
      Fraction of object selected: 100.00% (= 124770 points) ; Access speed: 1.74GB/sec
      Fraction of object retained:  90.79% (= 113280 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 19.3%; Transf. time frac.: 61.7%; Resulting pix preparation: 19.0%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 124799 pixels -- processing data... -----> retained 113303 pixels
-----
Number of points in input object: 124799
      Fraction of object selected: 100.00% (= 124799 points) ; Access speed: 1.78GB/sec
      Fraction of object retained:  90.79% (= 113303 points)

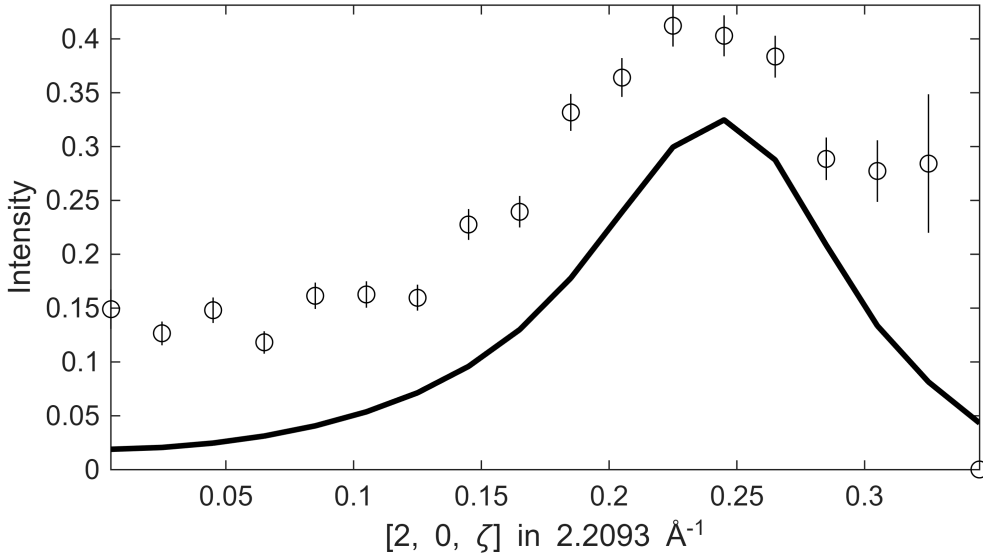
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 19.8%; Transf. time frac.: 60.6%; Resulting pix preparation: 19.5%
-----

```


Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $50 \leq E \leq 70$

$\zeta = -0.005:0.02:0.615$ in $[2, 0, \zeta]$



**** En = 70±10

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.173

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 131994 pixels -- processing data... -----> included 119578 pixels

Number of points in input object: 131994

Fraction of object selected: 100.00% (= 131994 points) ; Access speed: 1.91GB/sec

Fraction of object retained: 90.59% (= 119578 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 38.6%; Transf. time frac.: 56.5%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 131790 pixels -- processing data... -----> retained 119392 pixels

Number of points in input object: 131790

Fraction of object selected: 100.00% (= 131790 points) ; Access speed: 1.63GB/sec

Fraction of object retained: 90.59% (= 119392 points)

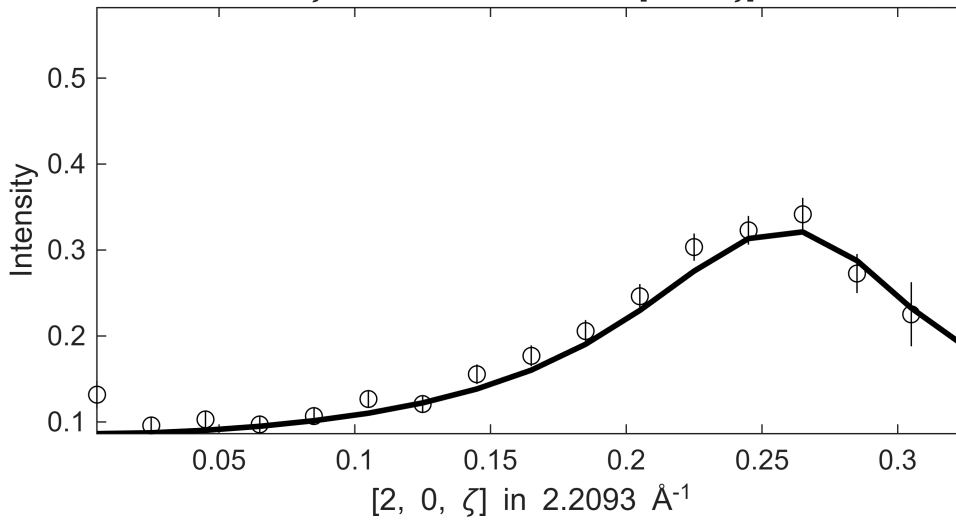
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 16.6%; Transf. time frac.: 63.6%; Resulting pix preparation: 19.8%

Fe_ei200_align_sym3.sqw

S=0.91±0.049; J0= 28±0.74; gamma=30.7433±0.944142

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 60 ≤ E ≤ 80
ζ=-0.005:0.02:0.615 in [2, 0, ζ]

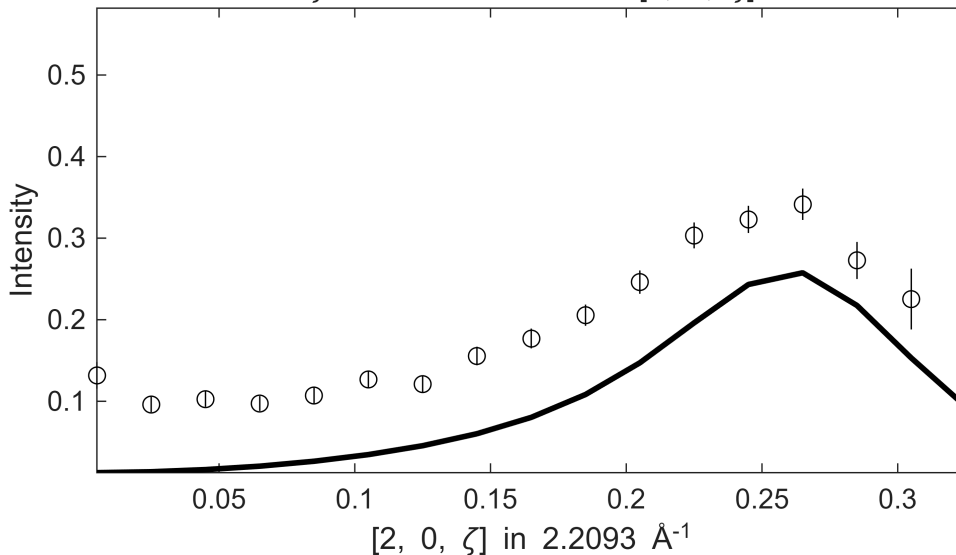


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 131994 pixels -- processing data... -----> retained 119578 pixels
-----
Number of points in input object: 131994
  Fraction of object selected: 100.00% (= 131994 points) ; Access speed: 1.32GB/sec
  Fraction of object retained: 90.59% (= 119578 points)

**** Total time in cut(sqw) : 0.0sec Including:
  Data access time fraction: 26.9%; Transf. time frac.: 55.2%; Resulting pix preparation: 17.8%
-----
```

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 60 ≤ E ≤ 80
ζ=-0.005:0.02:0.615 in [2, 0, ζ]



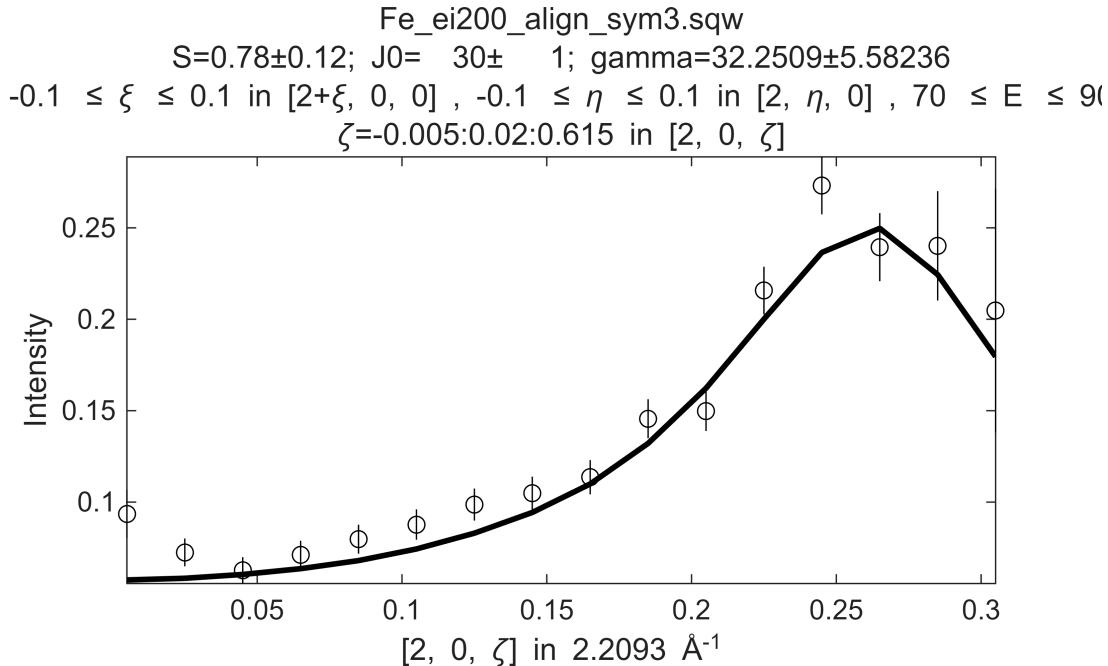
```
*****
**** En = 80±10
*****
Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.4663
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 136534 pixels -- processing data... -----> included 124607 pixels
-----
Number of points in input object: 136534
  Fraction of object selected: 100.00% (= 136534 points) ; Access speed: 1.84GB/sec
  Fraction of object retained: 91.26% (= 124607 points)
```

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 36.9%; Transf. time frac.: 57.3%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 136313 pixels -- processing data... -----> retained 124402 pixels
-----
Number of points in input object: 136313
      Fraction of object selected: 100.00% (= 136313 points) ; Access speed: 1.43GB/sec
      Fraction of object retained:  91.26% (= 124402 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 18.2%; Transf. time frac.: 51.0%; Resulting pix preparation: 30.8%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 136534 pixels -- processing data... -----> retained 124607 pixels
-----
Number of points in input object: 136534
      Fraction of object selected: 100.00% (= 136534 points) ; Access speed: 1.55GB/sec
      Fraction of object retained:  91.26% (= 124607 points)

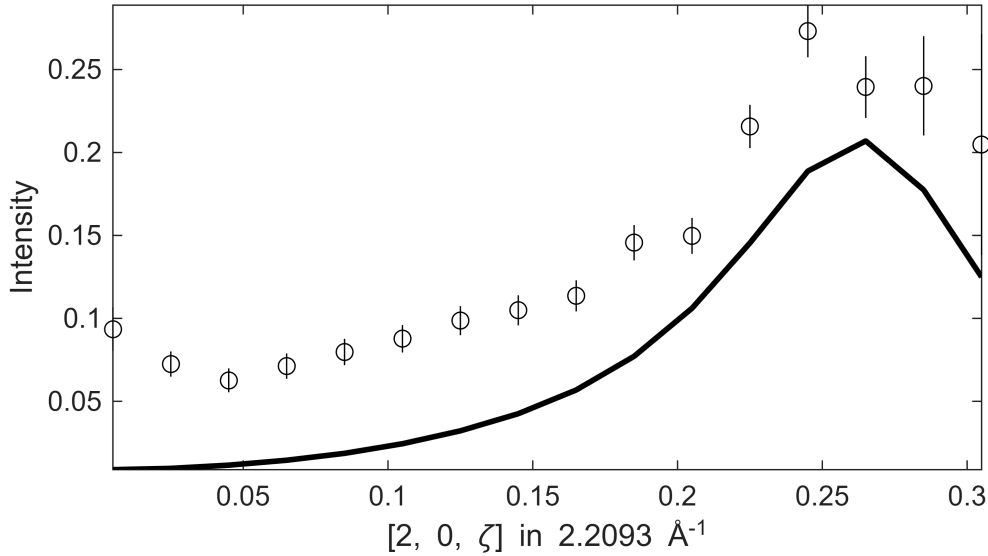
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 20.8%; Transf. time frac.: 57.2%; Resulting pix preparation: 21.9%
-----

```

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $70 \leq E \leq 90$

$\zeta = -0.005:0.02:0.615$ in $[2, 0, \zeta]$



**** En = 90±10

Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.0528

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 126639 pixels -- processing data... -----> included 115323 pixels

Number of points in input object: 126639

Fraction of object selected: 100.00% (= 126639 points) ; Access speed: 1.99GB/sec

Fraction of object retained: 91.06% (= 115323 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 33.2%; Transf. time frac.: 61.9%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 126529 pixels -- processing data... -----> retained 115232 pixels

Number of points in input object: 126529

Fraction of object selected: 100.00% (= 126529 points) ; Access speed: 1.72GB/sec

Fraction of object retained: 91.07% (= 115232 points)

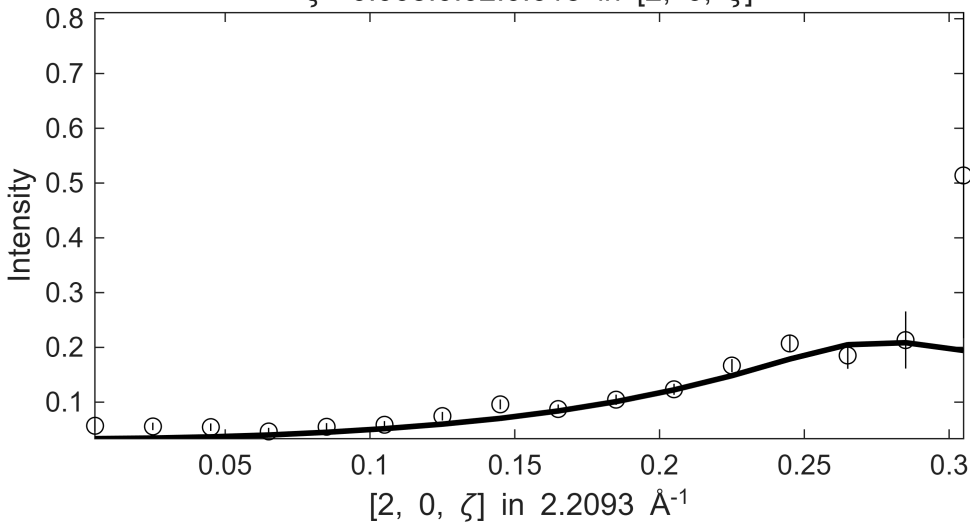
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 18.8%; Transf. time frac.: 63.3%; Resulting pix preparation: 17.9%

Fe_ei200_align_sym3.sqw

S= 1.1± 0.2; J0= 31± 3.4; gamma=52.6742±4.95133

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 80 ≤ E ≤ 10
ζ=-0.005:0.02:0.615 in [2, 0, ζ]



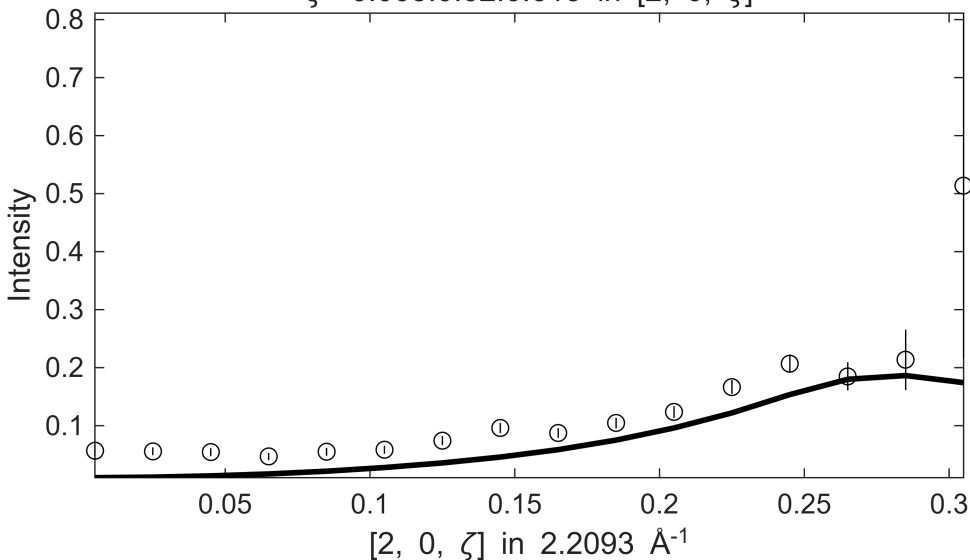
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 126639 pixels -- processing data... -----> retained 115323 pixels

Number of points in input object: 126639
Fraction of object selected: 100.00% (= 126639 points) ; Access speed: 1.80GB/sec
Fraction of object retained: 91.06% (= 115323 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 19.8%; Transf. time frac.: 60.1%; Resulting pix preparation: 20.0%

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 80 ≤ E ≤ 10
ζ=-0.005:0.02:0.615 in [2, 0, ζ]



**** En = 100±10

Warning: Dataset:26 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 7.6246
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 118733 pixels -- processing data... -----> included 103802 pixels

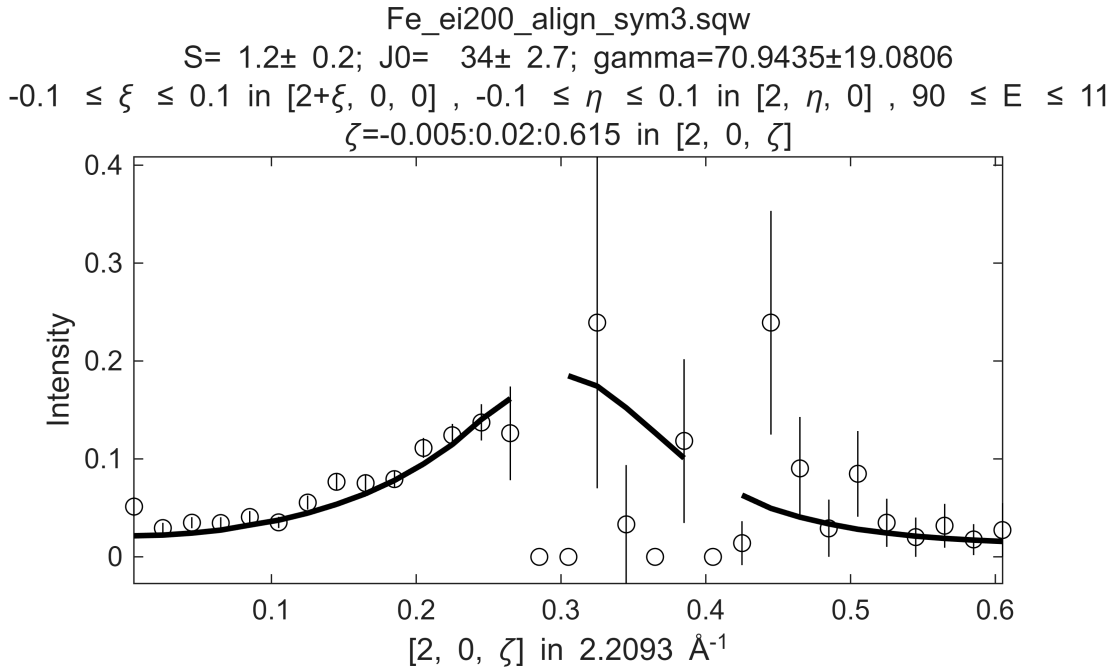
Number of points in input object: 118733
Fraction of object selected: 100.00% (= 118733 points) ; Access speed: 1.90GB/sec
Fraction of object retained: 87.42% (= 103802 points)

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 34.5%; Transf. time frac.: 59.2%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 116807 pixels -- processing data... -----> retained 102043 pixels
-----
Number of points in input object: 116807
      Fraction of object selected: 100.00% (= 116807 points) ; Access speed: 1.74GB/sec
      Fraction of object retained: 87.36% (= 102043 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 19.7%; Transf. time frac.: 62.2%; Resulting pix preparation: 18.0%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 118733 pixels -- processing data... -----> retained 103802 pixels
-----
Number of points in input object: 118733
      Fraction of object selected: 100.00% (= 118733 points) ; Access speed: 1.78GB/sec
      Fraction of object retained: 87.42% (= 103802 points)

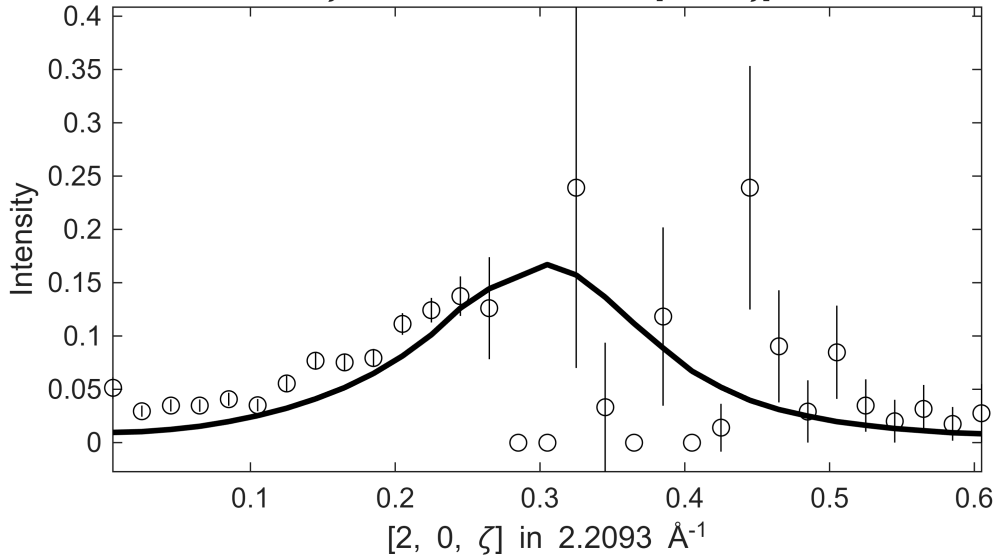
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 21.3%; Transf. time frac.: 58.4%; Resulting pix preparation: 20.3%
-----

```

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $90 \leq E \leq 11$

$\zeta = -0.005:0.02:0.615$ in $[2, 0, \zeta]$



**** En = 110±10

Warning: Dataset:26 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 7.6246

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 259314 pixels -- processing data... -----> included 224720 pixels

Number of points in input object: 259314

Fraction of object selected: 100.00% (= 259314 points) ; Access speed: 1.67GB/sec

Fraction of object retained: 86.66% (= 224720 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 41.8%; Transf. time frac.: 55.9%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 257265 pixels -- processing data... -----> retained 222671 pixels

Number of points in input object: 257265

Fraction of object selected: 100.00% (= 257265 points) ; Access speed: 1.65GB/sec

Fraction of object retained: 86.55% (= 222671 points)

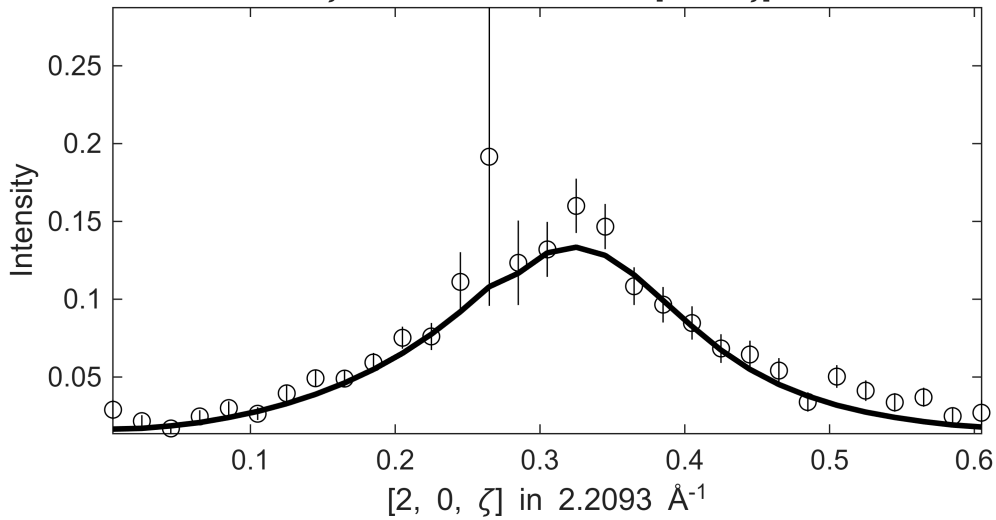
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 24.9%; Transf. time frac.: 58.5%; Resulting pix preparation: 16.5%

Fe_ei200_align_sym3.sqw

S= 1.1±0.13; J0= 35± 1.7; gamma=93.0224±15.4021

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 100 ≤ E ≤ 1;
ζ=-0.005:0.02:0.615 in [2, 0, ζ]

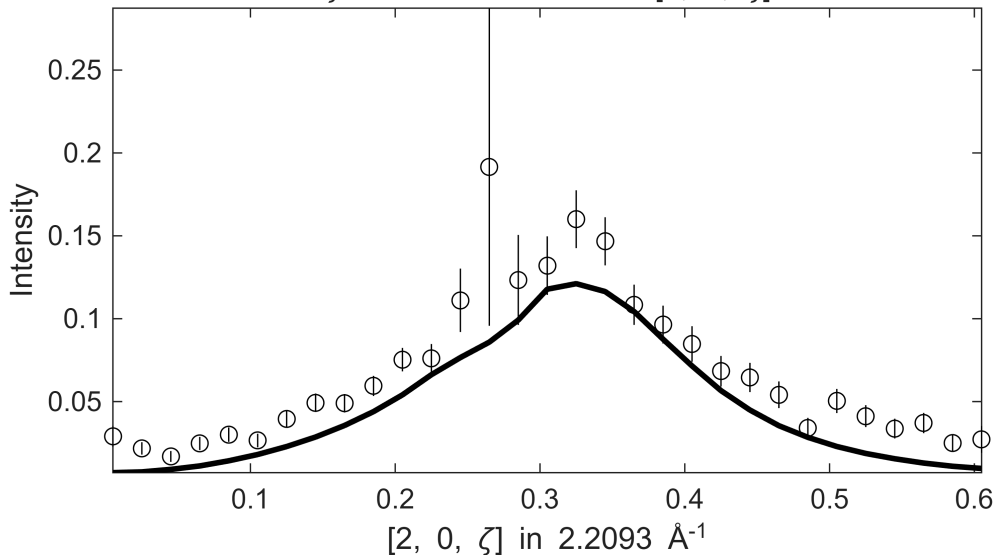


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 259314 pixels -- processing data... -----> retained 224720 pixels
-----
Number of points in input object: 259314
  Fraction of object selected: 100.00% (= 259314 points) ; Access speed: 1.65GB/sec
  Fraction of object retained: 86.66% (= 224720 points)

**** Total time in cut(sqw) : 0.0sec Including:
  Data access time fraction: 23.8%; Transf. time frac.: 60.5%; Resulting pix preparation: 15.7%
-----
```

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 100 ≤ E ≤ 1;
ζ=-0.005:0.02:0.615 in [2, 0, ζ]



```
*****
**** En = 120±10
*****
Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 2.0528
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 680386 pixels -- processing data... -----> included 611444 pixels
-----
Number of points in input object: 680386
  Fraction of object selected: 100.00% (= 680386 points) ; Access speed: 1.61GB/sec
  Fraction of object retained: 89.87% (= 611444 points)
```



```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction:  57.8%; Transf. time frac.:  40.6%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  679962 pixels -- processing data... -----> retained  611099 pixels
-----
Number of points in input object: 679962
      Fraction of object selected:  100.00% (=  679962 points) ; Access speed:      2.02GB/sec
      Fraction of object retained:   89.87% (=  611099 points)

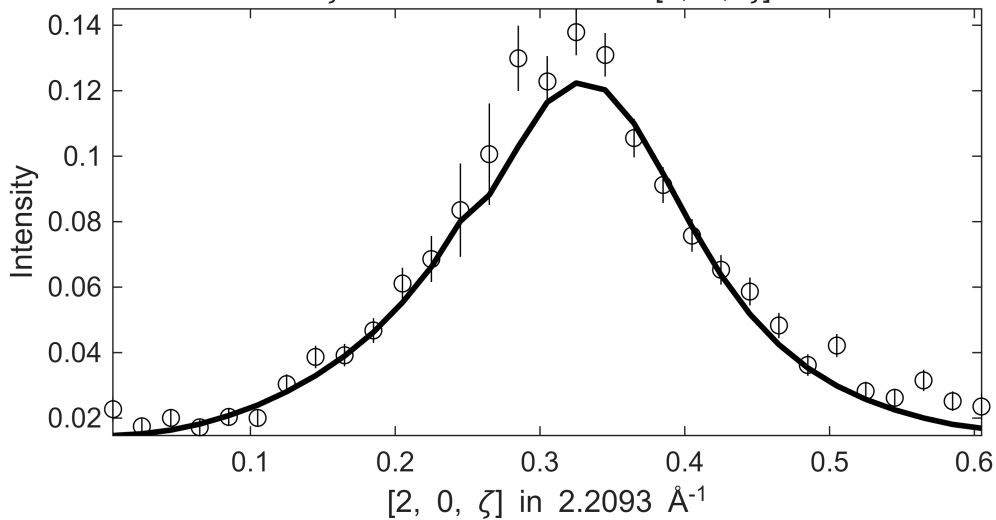
**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  28.3%; Transf. time frac.:  51.8%; Resulting pix preparation:  19.9%
-----

```

Fe_ei200_align_sym3.sqw

S= 1 ± 0.072 ; J0= 35 ± 0.8 ; gamma= 86.3837 ± 6.86645

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $110 \leq E \leq 115$ eV
 $\zeta = -0.005:0.02:0.615$ in $[2, 0, \zeta]$



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  680386 pixels -- processing data... -----> retained  611444 pixels
-----
Number of points in input object: 680386
      Fraction of object selected:  100.00% (=  680386 points) ; Access speed:      1.72GB/sec
      Fraction of object retained:   89.87% (=  611444 points)

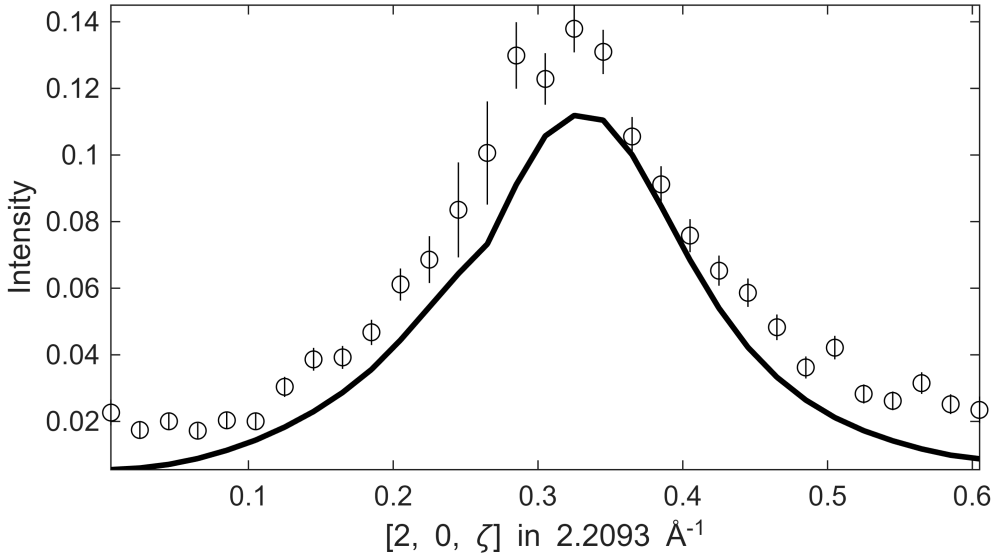
**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  31.2%; Transf. time frac.:  50.5%; Resulting pix preparation:  18.3%
-----

```

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $110 \leq E \leq 115$

$\zeta = -0.005:0.02:0.615$ in $[2, 0, \zeta]$



**** En = 130±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 1295075 pixels -- processing data... -----> included 1214206 pixels

Number of points in input object: 1295075

Fraction of object selected: 100.00% (= 1295075 points) ; Access speed: 2.07GB/sec

Fraction of object retained: 93.76% (= 1214206 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 56.4%; Transf. time frac.: 42.8%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 1295075 pixels -- processing data... -----> retained 1214206 pixels

Number of points in input object: 1295075

Fraction of object selected: 100.00% (= 1295075 points) ; Access speed: 1.60GB/sec

Fraction of object retained: 93.76% (= 1214206 points)

**** Total time in cut(sqw) : 0.2sec Including:

Data access time fraction: 32.2%; Transf. time frac.: 47.9%; Resulting pix preparation: 20.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 1295075 pixels -- processing data... -----> retained 1214206 pixels

Number of points in input object: 1295075

Fraction of object selected: 100.00% (= 1295075 points) ; Access speed: 1.76GB/sec

Fraction of object retained: 93.76% (= 1214206 points)

**** Total time in cut(sqw) : 0.2sec Including:

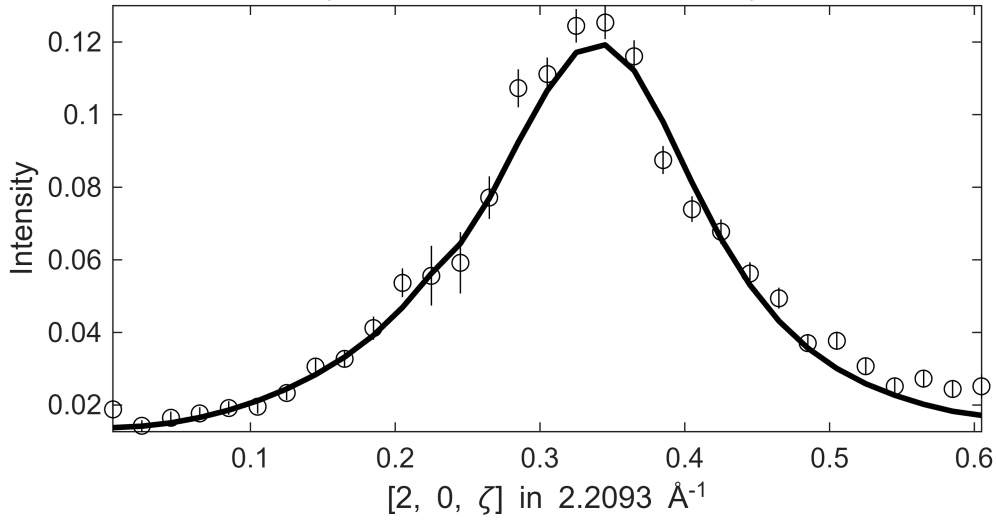
Data access time fraction: 32.3%; Transf. time frac.: 48.7%; Resulting pix preparation: 19.0%

Fe_ei200_align_sym3.sqw

S=0.93±0.054; J0= 35±0.66; gamma=77.5298±5.57022

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 120 ≤ E ≤ 140

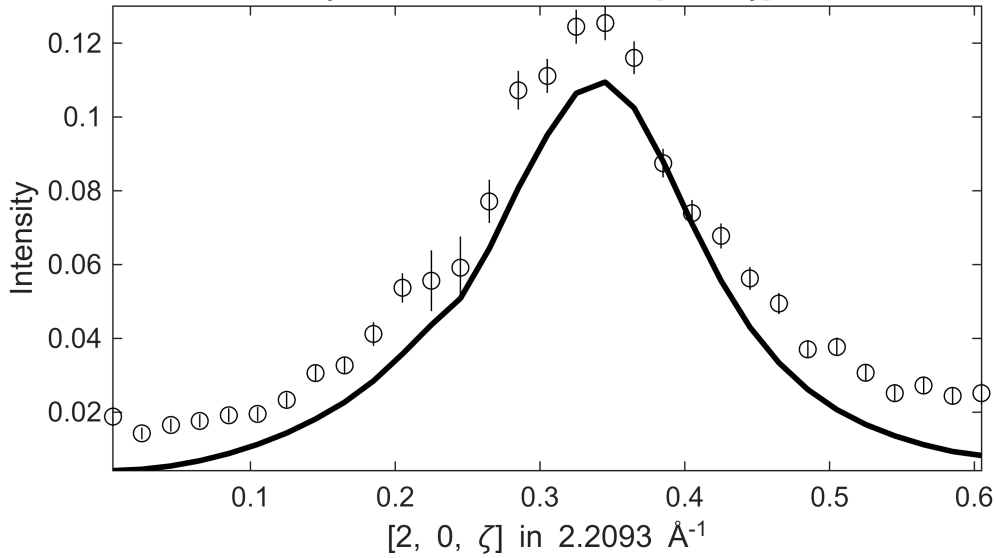
ζ=-0.005:0.02:0.615 in [2, 0, ζ]

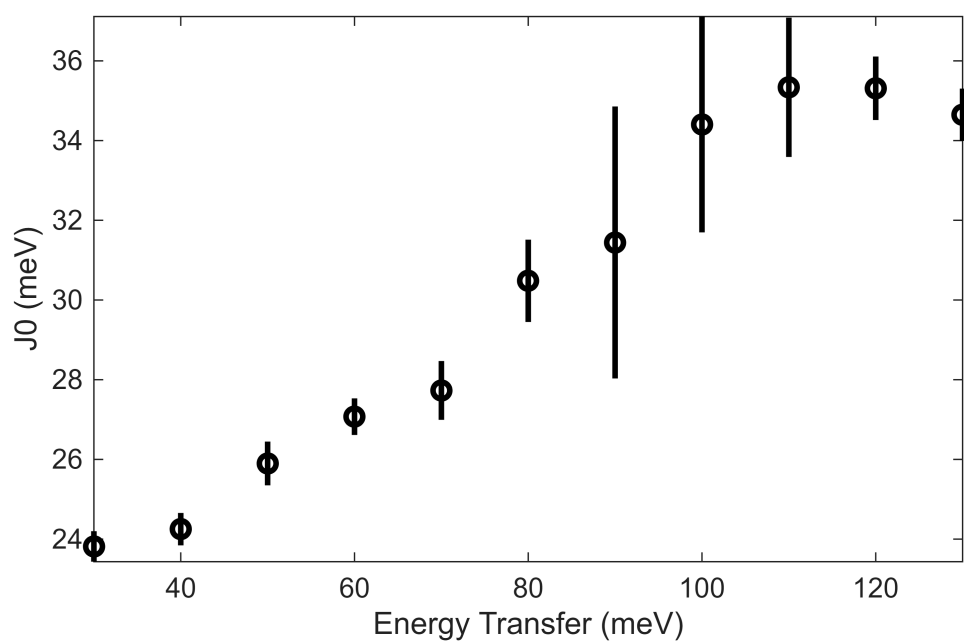
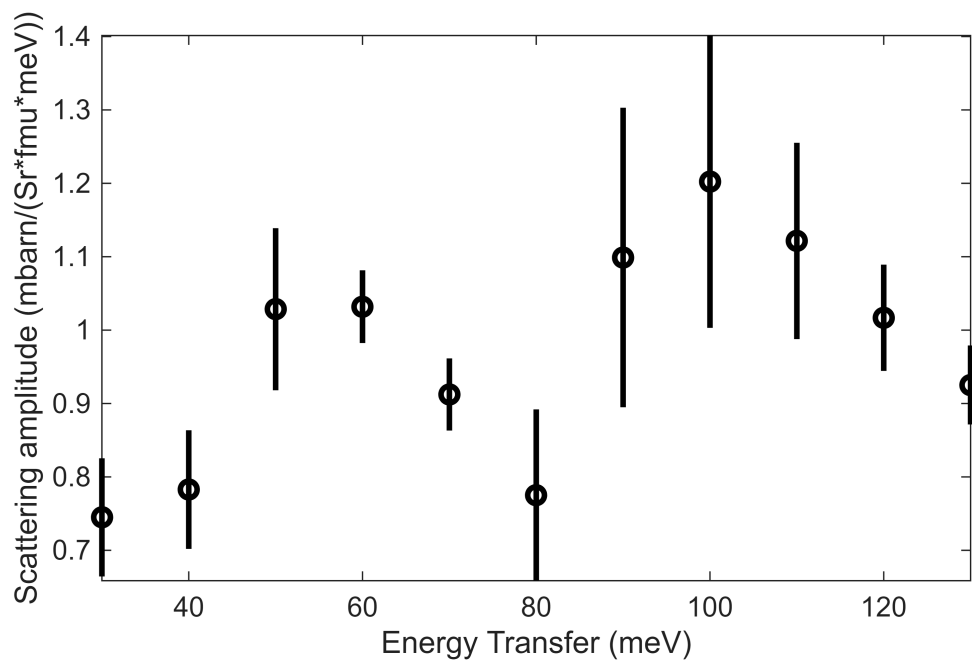


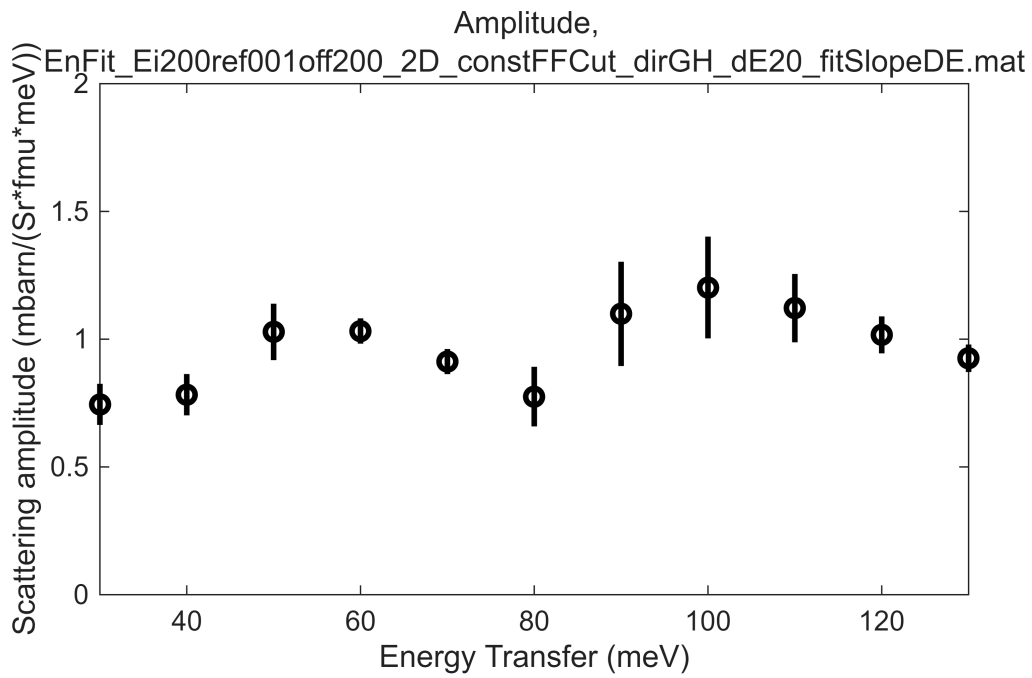
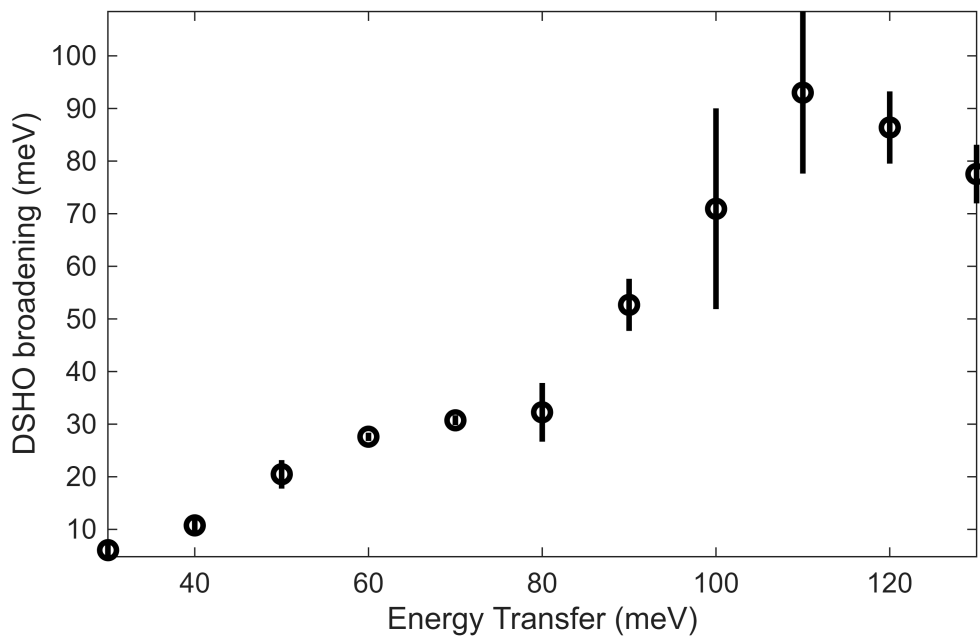
Fe_ei200_align_sym3.sqw

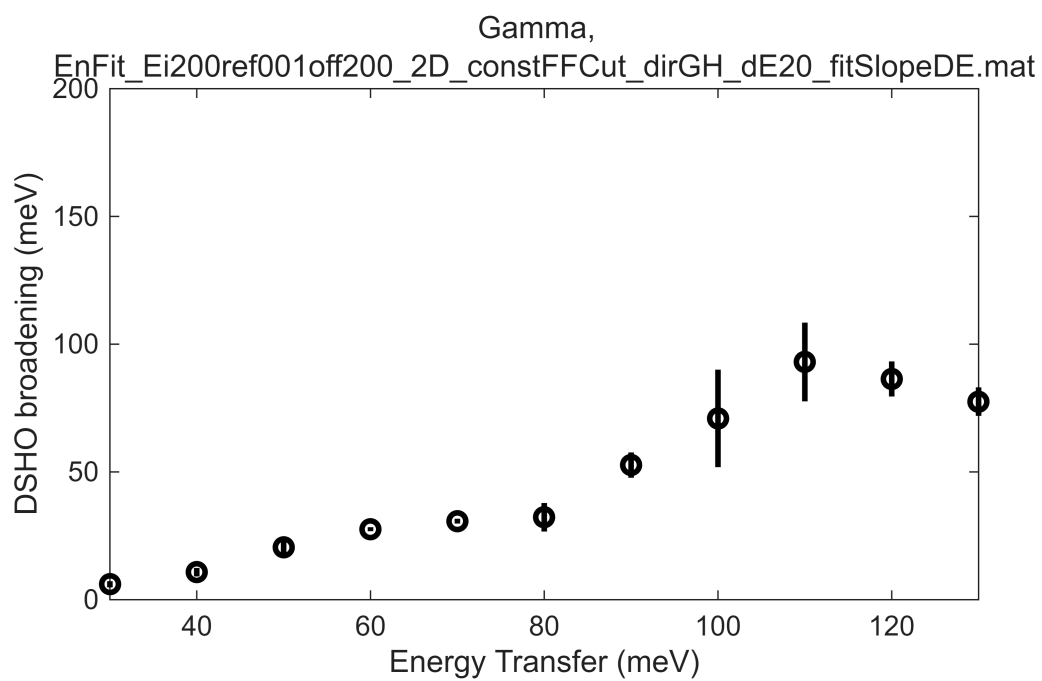
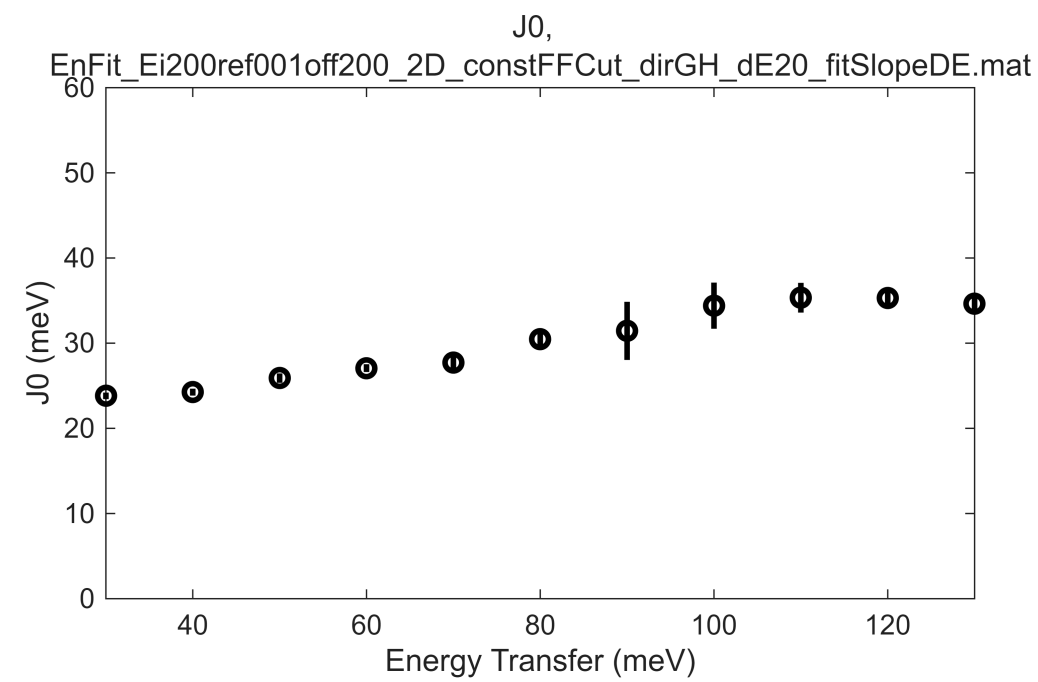
-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 120 ≤ E ≤ 140

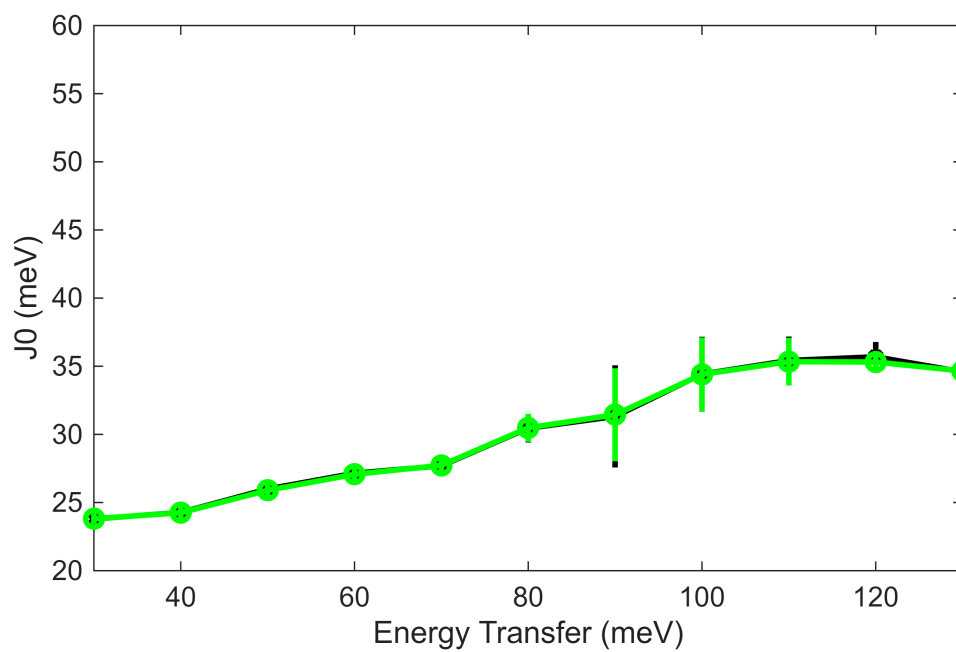
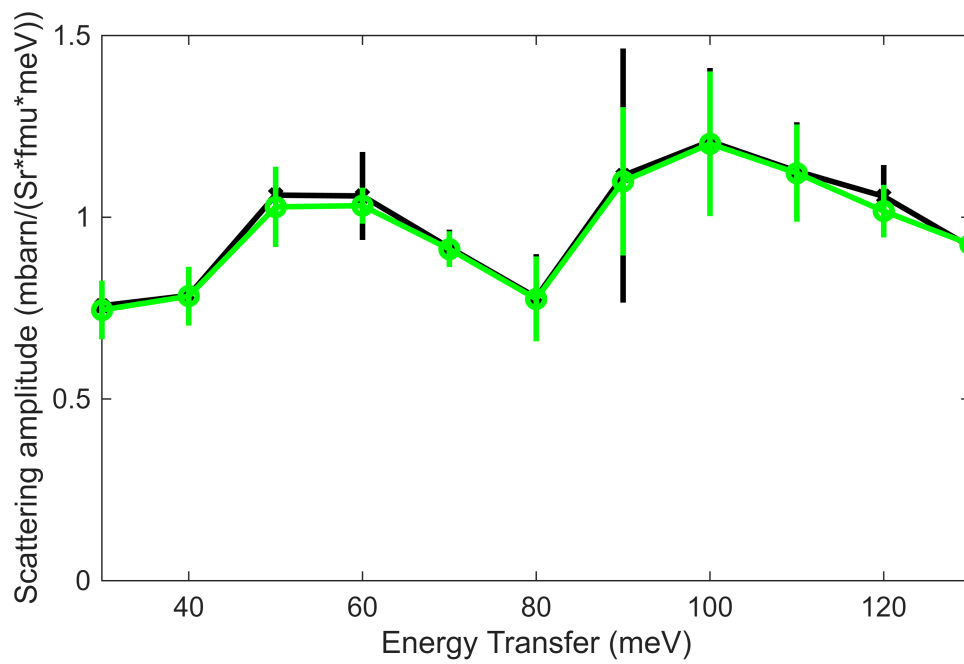
ζ=-0.005:0.02:0.615 in [2, 0, ζ]

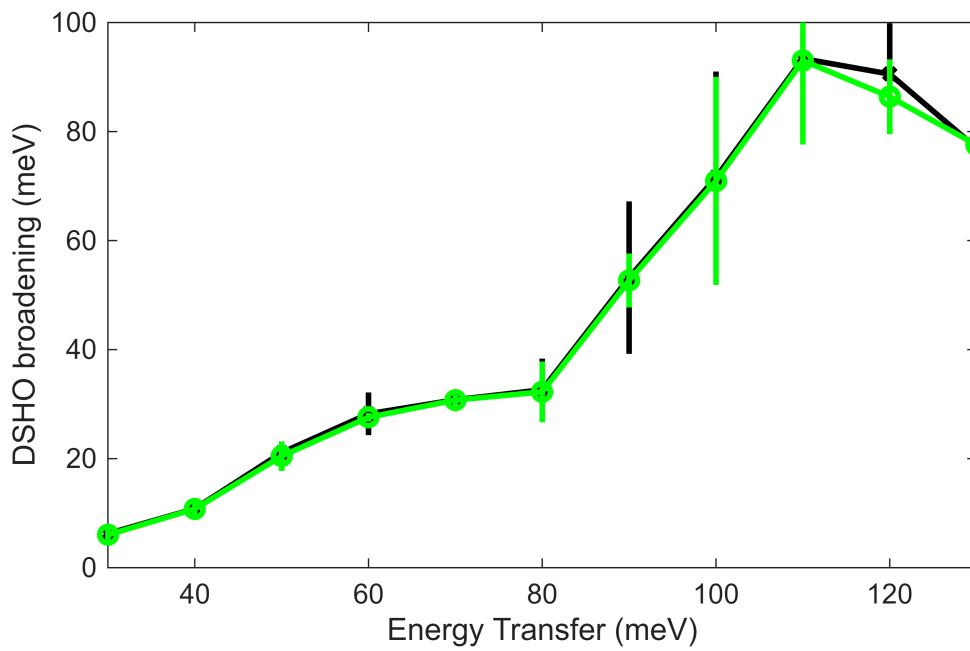






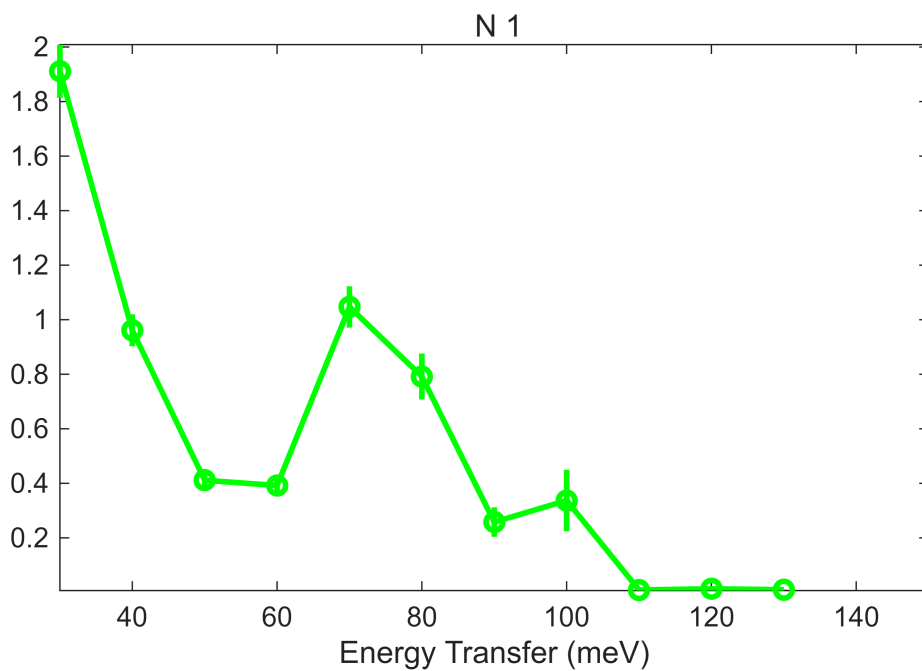




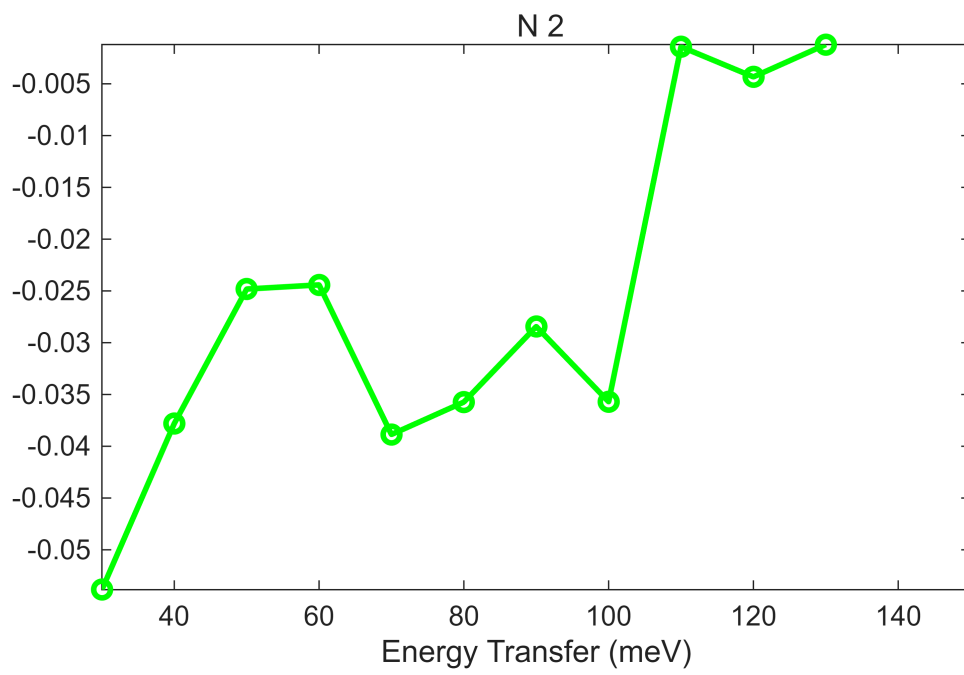


beep
beep
beep

```
bg_data = extract_bg_par(fit_res_200fix_bg.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;
```



```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure
```

```
fit_resEi200 = fit_res_200fix_bg;  
save(res_name, "fit_resEi200")
```